# VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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**A Synopsis on**
**"Automated Bug Report Processing System Using AI"**

**Submitted by**


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# ABSTRACT

The rapid growth of software applications and digital services has significantly increased the number of security vulnerabilities reported by testers and security teams. Managing and prioritizing these vulnerability reports manually is time-consuming, inefficient, and prone to inconsistencies. The proposed system, "AI-Based Vulnerability Triage in a Private Bug Bounty Platform," aims to develop an intelligent platform that assists organizations in automatically analyzing and prioritizing vulnerability reports using Machine Learning techniques. The system utilizes real-world vulnerability data obtained from the National Vulnerability Database and applies Natural Language Processing (NLP) and Naive Bayes classification to categorize reports into severity levels such as Critical, High, Medium, and Low. The platform also provides a private environment for internal testers to submit vulnerability reports securely. By automating the triage process, the proposed system reduces manual effort, improves vulnerability response time, and supports efficient cybersecurity management. The system is designed as a lightweight and scalable prototype demonstrating the practical application of Artificial Intelligence in vulnerability management workflows.

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# CHAPTER – 1

# INTRODUCTION

## 1.1 Aim

The main aim of the project is to develop an AI-based vulnerability triage system integrated into a bug bounty platform that can automatically analyze, classify, and prioritize vulnerability reports based on their severity. The system is intended to reduce manual effort involved in vulnerability management and improve the efficiency of security teams in identifying critical issues.

## 1.2 Overview

With the increasing dependence on software applications and online services, cybersecurity vulnerabilities have become a major concern for organizations. Security teams and testers continuously identify and report vulnerabilities present in applications, networks, and systems. However, managing large numbers of vulnerability reports manually is a difficult and time-consuming task.

Traditional bug tracking systems mainly focus on storing reports and require human analysts to manually review and prioritize each vulnerability. This process often leads to delays, inconsistent prioritization, and increased workload for security teams.

The proposed system introduces an AI-based vulnerability triage mechanism that automatically analyzes vulnerability reports using Machine Learning and Natural Language Processing techniques. The system predicts the severity level of reported vulnerabilities and prioritizes them accordingly. The platform is designed as a bug bounty environment where authorized internal testers can securely submit vulnerability reports.

The project utilizes real-world cybersecurity data from the National Vulnerability Database to train the Machine Learning model and improve classification performance.

## 1.3 Problem Statement

The increasing number of vulnerability reports generated by internal testers and

security teams has made manual vulnerability triage inefficient, time-consuming, and difficult to scale. Existing systems largely depend on human analysts to review, classify, and prioritize reports, resulting in delayed identification of critical vulnerabilities, inconsistent prioritization, duplicate report handling issues, and increased operational workload. Hence, there is a need for an AI-based vulnerability triage system capable of automatically analyzing and prioritizing vulnerability reports to support faster and more efficient cybersecurity decision-making.

## 1.4 Existing and Proposed System

### Existing System

Existing bug tracking and vulnerability management platforms such as:

- BugBase
- Bugtroope

mainly depend on manual review and prioritization by security analysts. Although these systems provide vulnerability management functionality, they often lack lightweight AI-assisted automated triage mechanisms for severity prediction and prioritization.

**Limitations of Existing System**

- Manual prioritization process
- Increased analysis time
- High dependency on security experts
- Difficulty handling large-scale reports
- Delayed response to critical vulnerabilities

### Proposed System

The proposed system introduces an AI-based vulnerability triage platform integrated into a private bug bounty environment. The system allows authorized internal testers to submit vulnerability reports through a secure interface.

The submitted reports are processed using Natural Language Processing techniques and analyzed using a Machine Learning model based on the Naive Bayes algorithm. The system automatically predicts the severity level of vulnerabilities and prioritizes them accordingly.

**Features of Proposed System**

- AI-based vulnerability severity classification.
- Automated prioritization of reports.
- Secure bug bounty environment.
- Reduced manual effort.
- Faster vulnerability management workflow.
- Real-world dataset integration using NVD data.

## 1.5 SDG MAPPING

The proposed project contributes to multiple Sustainable Development Goals (SDGs) by promoting secure digital infrastructure, technological innovation, and responsible cybersecurity practices.

**SDG 9 – Industry, Innovation and Infrastructure**

The project supports innovation in cybersecurity management through Artificial Intelligence and Machine Learning technologies. The system strengthens digital infrastructure security by improving vulnerability management and prioritization processes.

**SDG 8 – Decent Work and Economic Growth**

Cybersecurity incidents can cause major financial and operational losses for organizations. The proposed system improves vulnerability response efficiency and helps organizations reduce security risks, contributing to stable and secure digital operations.

**SDG 16 – Peace, Justice and Strong Institutions**

The project contributes toward strengthening digital trust and cybersecurity practices by assisting organizations in managing vulnerabilities effectively and protecting systems from cyber threats.

## 1.5Hardware and Software Requirements

## Hardware Requirements

| Category | Requirements |
|---|---|
| Processor | Intel Core i5 or above |
| RAM | Minimum 8 GB |
| Storage | 256 GB SSD or above |
| Operating System | Windows 10 / Linux |

## Software Requirements

| Programming Language | Python |
|---|---|
| Web Framework | Streamlit |
| Database | SQLite |
| Machine Learning Library | scikit-learn |
| Data Processing Libraries | Pandas, NumPy |
| Dataset Source | National Vulnerability Database |

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